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Informational faultlines, team boundary work, and idea implementation: The moderating role of paradoxical leadership

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ABSTRACT

Drawing on social network theory, our study develops a theoretical model delineating the impact of informational faultlines, paradoxical leadership, and team boundary work on idea implementation. Drawing from empirical research using a dataset of 82 teams from multi-source and multi-timepoints, our findings reveal that informational faultlines positively influence team boundary work (boundary spanning, buffering, and reinforcement) and idea implementation. Team boundary work (boundary spanning, buffering, and reinforcement) mediates the relationship between informational faultlines and idea implementation. Paradoxical leadership moderates the mediating effect of boundary buffering and spanning between informational faultlines and idea implementation. The findings of our study provide theoretical insights and practical implications for the management of innovative teams in businesses, team establishment, and the recruitment and selection of leaders.

1. Introduction

In today's dynamic and competitive environment, organizations must continuously enhance their innovation capabilities to stay relevant (Wei et al., 2023). While generating creative ideas is essential, the ability to transform these ideas into tangible products, services, or methods with commercial value has become a cornerstone of sustainable competitive advantage (Kim & Choi, 2023). To achieve this, organizations increasingly rely on diverse, knowledge-based teams to execute idea implementation tasks (Somech & Drach-Zahavy, 2013). However, team diversity presents both opportunities and challenges. While it provides access to heterogeneous knowledge and resources, it can also reduce trust and cohesion, thereby undermining collaboration efficiency during the implementation process (Bezrukova et al., 2012).

Informational faultlines refer to the hypothetical dividing lines within teams that arise from the alignment of members' attributes (e.g., functional areas, education, tenure), offering a nuanced perspective on these challenges (Lau & Murnighan, 1998). Unlike traditional diversity measures, which treat it as an aggregate construct, informational faultlines capture the clustering of attributes into subgroups, revealing the social dynamics that can shape team interactions (Bezrukova et al., 2009). These faultlines are often viewed as sources of division, but they can also foster positive outcomes. Subgroups formed along informational faultlines can serve as hubs of unique knowledge and expertise, promoting specialization and offering diverse perspectives that, when integrated effectively, enhance creativity and resource utilization (Yao et al., 2021). Thus, understanding the impact of informational faultlines on idea implementation is critical for managing diverse teams and fostering innovation. Our study addresses this need by exploring the mechanisms through which these faultlines influence idea implementation and identifying the conditions under which they have the most significant effects.

A critical factor in leveraging the potential of informational faultlines for idea implementation is team boundary work. Team boundary work, which encompasses both inward-focused processes (strengthening internal collaboration and cohesion) and outward-focused efforts (engaging external stakeholders and acquiring necessary resources), offers a dynamic framework for addressing the challenges posed by faultlines. This dual focus enables teams to transcend subgroup divides, integrate diverse perspectives, and align internal dynamics with external demands to foster effective idea implementation (Faraj & Yan, 2009; Ozen & Ozturk-Kose, 2023). Despite its potential, existing research has largely overlooked the systemic role of team boundary work in mediating the effects of informational faultlines on idea implementation.

Additionally, the changing others' relationships framework of social network theory suggests that paradoxical leadership can guide teams with informational faultlines in balancing the competing demands of internal collaboration and external resource acquisition. This, in turn,

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enhances their capacity to implement innovative ideas (Halevy et al., 2019). Teams with informational faultlines face multiple conflicting demands, such as the need to strengthen relationship capital internally while acquiring resources and support externally (Qu & Liu, 2017). Paradoxical leadership not only reduces disagreements among members due to cognitive differences by establishing effective communication bridges but also guides members to examine and balance the competitive and complementary inward and outward boundary work from a dialectical perspective (Volk et al., 2023; Zhang et al., 2015). This fosters mutual psychological identification and mutual trust among members, forming cohesion, and ultimately driving the team to translate more creativity into corresponding products, services, or processes (Valtonen et al., 2023; Wei et al., 2023). However, existing research largely overlooks the moderating influence of paradoxical leadership on the relationship between informational faultlines, team boundary work, and idea implementation.

Our study introduces several innovations to the current research in the following areas. First, it develops a theoretical model assessing the impact of informational faultlines, paradoxical leadership, and team boundary work on idea implementation, thus initiating the first exploration of the relationship between informational faultlines and idea implementation. Simultaneously, it extends the application of social network theory to the study of informational faultlines, offering new theoretical frameworks and perspectives for understanding the mechanisms underlying informational faultlines effects. Second, it introduces team boundary work into the study of the relationship between informational faultlines and idea implementation from an open system perspective, identifying boundary reinforcement, buffering, and spanning as key mediators. Third, it reveals the moderating role of paradoxical leadership in the "informational faultlines—team boundary work—idea implementation" relationship, offering a novel perspective on the boundary conditions that affect how informational faultlines influence idea implementation.

2. Theory and hypotheses

2.1. Theoretical background

2.1.1. Informational faultlines and idea implementation

Lau and Murnighan (1998) introduced the concept of team faultlines, defined as potential divisions among team members based on one or more attributes. Subsequent scholars have expanded on this foundational concept, positing that not only readily identifiable demographic traits, such as age or gender, can trigger faultlines, but also differences in education level, professional skills, and years of experience contribute to these faultlines (Carton & Cummings, 2012). Drawing on existing literature, our study defines informational faultlines as hypothetical divisions within teams, based on the alignment of multiple task-related demographic attributes, thereby segmenting teams into subgroups that are heterogeneous yet relatively homogeneous in knowledge (Bezrukova et al., 2009). In contrast to social faultlines, informational faultlines are more directly linked to work-related attributes and behavioral performance of team members, exerting a greater impact on the team innovation process (Qu & Liu, 2017). Team idea implementation, a core component of the team innovation process, primarily involves the systematic transformation of innovative ideas or concepts into tangible products, services, or processes (Somech & Drach-Zahavy, 2013). The core challenge of idea implementation lies in effectively integrating various resources, fostering collaboration and coordination among team members, and overcoming numerous obstacles encountered during the implementation phase (Kim & Choi, 2022). Accordingly, it can be seen that effective management of informational faultlines can enhance creative collaboration within teams while poor management may lead to increased conflict and reduced collaboration, thereby negatively impacting idea implementation (Yao et al., 2021).

2.1.2. Team boundary work

Team boundary work refers to the process through which teams effectively manage the exchange of resources between themselves and their external environment to optimize resource allocation and successfully achieve team goals, encompassing both inward and outward boundary work (Faraj & Yan, 2009; Yan & Louis, 1999). Inward boundary work, also known as boundary reinforcement, involves consolidating team boundaries to ensure uniqueness and superiority, thereby gaining a competitive advantage. Boundary reinforcement is facilitated through the explicit demarcation of a team's internal and external boundaries, which instills a strong sense of identity and belonging among team members. Outward boundary work includes both boundary spanning and boundary buffering. Boundary spanning emphasizes the need for team members to actively communicate, coordinate, and collaborate with the external environment, acquiring key resources and external support through an open approach to enhance competitive capabilities and market adaptability. Boundary buffering involves teams implementing strategies and measures to protect themselves from external instabilities, ensuring that the team can maintain internal stability and operate efficiently in a changing environment (Faraj & Yan, 2009). This suggests that effective management of team boundaries plays a crucial role in fostering idea implementation within informational faultlines.

2.1.3. Paradoxical leadership

The essence of paradoxical leadership lies in exhibiting contradictory leadership behaviors and traits to manage the irreconcilable paradoxes within teams, thereby achieving a leadership style that simultaneously satisfies the needs for organizational structure and employee individualization (Zhang et al., 2022; Zhang et al., 2015). Paradoxical leadership encompasses five dimensions, including maintaining close relationships while keeping an appropriate distance, treating subordinates fairly while respecting individual differences, and others. Paradoxical leadership, particularly suited to rapidly changing and highly uncertain organizational environments, fosters a proactive and innovative spirit among employees as they face market challenges, thereby driving sustained development in complex and competitive situations (Zhang et al., 2015). This shows that paradoxical leadership helps balance internal cohesion and external collaboration needs, enabling teams to fully leverage the diversity brought about by informational faultlines, thereby facilitating idea implementation.

2.2. Theoretical framework

Social network theory views the social network system as a whole to explain social behavior by primarily exploring the relationships between social actors and their impact on social structures and behaviors (Uzzi, 1997). Social network theory posits that embedding mechanisms are based on mutual trust among members, offering theoretical support for the relationships between informational faultlines, team boundary work, and idea implementation (Zhang & Guler, 2020). On one hand, informational faultlines significantly broaden the team's embedding breadth through outward boundary management mechanisms, providing support and assistance for idea implementation from the team's operational environment, while extensive interaction with the external environment facilitates the dissemination, acceptance, and application of ideas across diverse groups, thereby enhancing the team innovation (Liu et al., 2021). Furthermore, informational faultlines manage and filter ineffective external information and resource inflows through team boundary buffering, thereby mitigating potential threats to idea implementation from external uncertainties (Faraj & Yan, 2009). On the other hand, informational faultlines expand the team's embedding depth through inward boundary work, thereby facilitating members' access to assistance and emotional support from colleagues within subgroups, creating favorable conditions for the idea implementation (Qu & Liu, 2017). Thus, team boundary work serves as a key mechanism through which informational faultlines influence idea implementation.

The changing others' relationships framework within social network theory emphasizes that team leadership, as a critical situational factor, can guide and influence members' attitudes and behaviors, as well as team processes and outputs (Halevy et al., 2019). Paradoxical leadership, by employing paradoxical thinking, plays a synergistic role in integrating contradictions, enabling informational faultlines to both strengthen internal team cohesion and leverage external boundary resources and support, ultimately facilitating smooth idea implementation and organizational innovation capability development (Volk et al., 2023; Zhang et al., 2015). Thus, paradoxical leadership moderates the "informational faultlines—team boundary work (boundary spanning, buffering, and reinforcement)—idea implementation" action chain. In summary, our study develops a conceptual model grounded in social network theory, demonstrating the impact of informational faultlines on idea implementation through team boundary work under the moderating effect of paradoxical leadership (see Fig. 1).

2.3. Hypotheses development

2.3.1. Informational faultlines and idea implementation

While informational faultlines may initiate social categorization and lead to negative consequences such as bias and trust deficits among subgroups (Bezrukova et al., 2012), our study, grounded in social network theory, posits that the diversified information resource base created by informational faultlines serves not only as a platform for facilitating information exchange and sharing but also as a pivotal force propelling idea implementation (Qu & Liu, 2017; Uzzi, 1997). For example, in cross-functional teams, the exchange of information and the clash of ideas between technical experts, marketing personnel, and product designers can foster interdisciplinary innovative solutions, significantly advancing product development and the creative realization of marketing strategies (Bezrukova et al., 2009). Compared to team faultlines based on differences in identity or values, informational faultlines have a more direct and significant influence on a team's ability to implement ideas because their characteristics enable team members to fully utilize each other's expertise and resources under task and goal orientations, effectively solving complex problems and achieving innovative objectives (Ma et al., 2022). Furthermore, team members within informational faultlines, possessing diverse information and professional backgrounds, are more likely to critically challenge and question others' viewpoints and proposals, effectively preventing mental inertia and blind consensus within the team and thereby ensuring the quality and feasibility of idea implementation (Cooper et al., 2014). Additionally, team members with diverse informational backgrounds strive to improve their argumentation and presentation skills in the process of advocating for their viewpoints, which not only enhances the depth and breadth of information but may also introduce new perspectives or technical methods (Yao et al., 2021), fostering healthy competition that strengthens team cohesion and maintain efficiency and motivation during the idea implementation phase (Baer, 2012). Based on the above discussion, we propose the following hypothesis:

Hypothesis 1 Informational faultlines positively influence idea implementation.

2.3.2. Informational faultlines and team boundary work

Drawing on social network theory, the positive influence of informational faultlines on boundary reinforcement is primarily manifested in several aspects. First, informational faultlines shape a clear internal structure within the team by creating subgroups based on similar educational backgrounds, years of work experience, and functional roles (Liu et al., 2021). These homogeneous subgroups can quickly build strong trust relationships, effectively reduce communication barriers, and promote unified cognition and alignment of goals among team members (Qu & Liu, 2017). Simultaneously, each subgroup's clearly defined responsibility and role within the overall team objectives and tasks significantly enhance internal cohesion and strengthen the team's unity and coordination during external presentations, thereby highlighting the team's distinctive advantages in a competitive market (Perry-Smith & Mannucci, 2017). Second, each informational subgroup focuses on optimizing its core capabilities, reducing disturbances from frequent external uncertainties, and helping the team accumulate expertise and skills in specific areas, thereby boosting competitiveness in those fields by concentrating on its distinct strengths (Feng et al., 2024). Moreover, clear boundaries between subgroups also help maintain the uniqueness of team culture and team member identity, further consolidating the effect of boundary reinforcement (Zhang & Guler, 2020). Finally, informational faultlines enable team management to monitor and allocate resources effectively according to the specific needs and strengths of each subgroup, thus maximizing resource utilization efficiency (Vandebeek et al., 2024). This not only strengthens team boundaries but also ensures that internal resource flow and distribution align with the overall team strategy and objectives (Leicht-Deobald et al., 2022). Based on the aforementioned analysis, we propose Hypothesis 2a:

Hypothesis 2a Informational faultlines positively influence boundary reinforcement.

Informational faultlines not only strengthen internal team boundaries but also significantly enhance boundary buffering in the face of external changes. Firstly, each informational subgroup focuses on processing information related to its field of expertise, effectively reducing information processing redundancy and delays (Yao et al., 2021). *This not only strengthens internal coordination and response speed but also enables rapid integration of information and resources to formulate strategies in response to external uncertainties and pressures, thereby effectively mitigating external influences* (Ma et al., 2022). *Secondly, informational subgroups deepen expertise and*

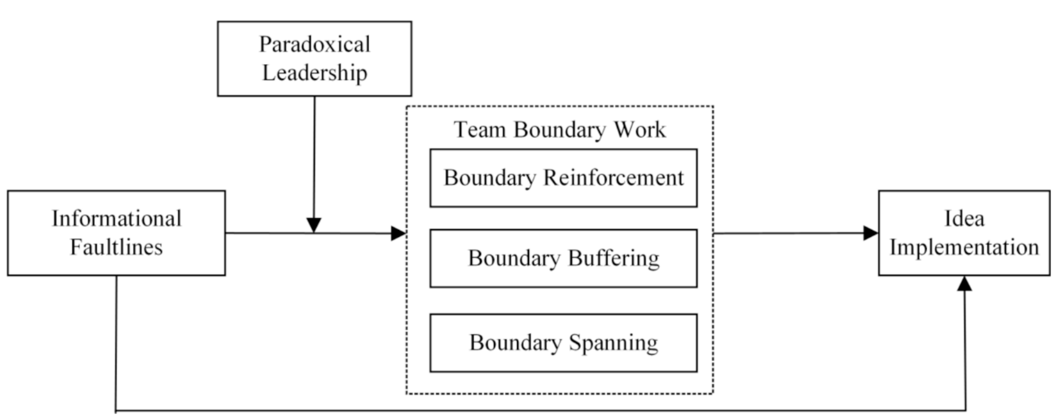

Fig. 1. Theoretical Research Model.

accumulate technology within their respective fields, creating unique knowledge bases and response strategies (Qu & Liu, 2017). *This not only heightens the team's sensitivity to changes within their professional areas but also enables them to devise more precise and effective strategies for specific external challenges* (Feng et al., 2024). *Furthermore, subgroups formed based on common professional backgrounds and skills are more likely to reach consensus, enhancing trust and support among members, allowing the team to maintain unified action and strategy when faced with external shocks, and effectively reducing the divergence and confusion caused by external instabilities* (Zhang & Guler, 2020). *Moreover, clear boundaries and professional orientations of each subgroup allow for more precise resource allocation to key areas and make risk management more targeted and effective* (Vandebeek et al., 2024). *When external environments change, the team can flexibly adjust resource distribution and risk control strategies based on the characteristics of each subgroup and the specific challenges faced, to maximize protection against adverse influences* (Ma et al., 2022). *Based on the above analysis, we propose Hypothesis* 2b*:*

Hypothesis 2b Informational faultlines positively influence boundary buffering.

Informational faultlines not only strengthen the internal structure of teams but also positively influence boundary spanning. Firstly, by forming specialized subgroups, informational faultlines facilitate deep development in specific areas, laying a solid foundation for team members to communicate and collaborate with external professional fields (Qu & Liu, 2017). *As each subgroup's expertise is enhanced, it is more likely to be recognized as a professional team by external partners, thus attracting external resources and support more easily* (Delios et al., 2024; Liu et al., 2021). *Secondly, members of each subgroup, being well-versed in their respective areas of expertise, can confidently and precisely showcase their team's capabilities and needs during external interactions, aiding in the efficient expression of needs, accurate assessment of collaboration opportunities, and ultimately optimizing resource allocation* (Zhang & Guler, 2020). *Thirdly, the extensive external networks created by informational faultlines enable different subgroups to engage in robust exchanges and integration of heterogeneous information and resources during interactions with the external environment, thereby stimulating innovative thinking and enhancing the team's adaptability and flexibility in uncertain conditions* (Qu & Liu, 2017). *Finally, informational faultlines allow for more precise strategic deployment during external communications and collaborations* (Vandebeek et al., 2024). *Depending on various external collaboration opportunities, the team can flexibly adjust its collaboration strategies and resource investments based on the characteristics of each subgroup and external environmental demands, thereby improving cooperation efficiency and outcomes* (Feng et al., 2024). *Based on the above analysis, we propose Hypothesis* 2c*:*

Hypothesis 2c Informational faultlines positively influence boundary spanning.

2.3.3. The mediating role of team boundary work

Informational faultlines enhance boundary reinforcement within teams by shaping members' awareness of their roles and boundaries. This process encourages members to collaborate more closely around shared team goals, thereby facilitating idea implementation (Bertello et al., 2022). Firstly, boundary reinforcement highlights the distinctiveness and superiority of the team by creating a supportive team atmosphere, enabling each member to recognize their value and the mutual respect and trust within the team (Langley et al., 2019). This enhances the functionality of team resource pooling and ultimately improves the efficiency of resource utilization, providing a stable and sustainable environment for idea implementation (De Vries et al., 2022). Secondly, boundary reinforcement, by establishing a shared vision and goals, provides members with a clear direction, transforming loosely connected members into a cohesive team with a focused drive (Faraj & Yan, 2009). This alignment directs the team's attention and resources toward accomplishing collective tasks, thereby stimulating intrinsic motivation, enhancing team efficiency, and strengthening team cohesion, ultimately facilitating successful idea implementation (Zhang & Guler, 2020). Finally, clearly defined team boundaries enable members to understand each other's roles and responsibilities more clearly, reducing the likelihood of conflicts and enhancing communication and collaboration efficiency which enhances the team's ability to leverage diverse strengths and elevate idea implementation (Langley et al., 2019). Building on the discussion in Hypothesis 2a, informational faultlines strengthen team boundaries through clear team structuring, enhanced cohesion, and efficient resource allocation, thereby facilitating conditions conducive to idea implementation. Therefore, we propose Hypothesis 3a:

Hypothesis 3a Informational faultlines indirectly positively influence idea implementation through the mediating role of boundary reinforcement.

Informational faultlines facilitate boundary buffering between the internal team environment and external influences, optimizing resource allocation and process management, and effectively promoting idea implementation. Firstly, boundary buffering maintains a closed and autonomous environment during the promotion of creative initiatives, allowing members to progress projects more freely and rationally without external disturbances, enabling close cooperation across departments and teams, and ultimately advancing idea implementation (Leicht Deobald et al., 2022). *Secondly, boundary buffering plays a crucial role in managing the inflow of external resources, ensuring the moderate introduction and high-quality selection of resources. The buffer zone acts as a filtering mechanism allowing only those external resources that meet organizational innovation needs and quality standards to enter, thus preventing the disruption of internal workflows due to resource surpluses or inefficient resource introduction* (Faraj & Yan, 2009). *Finally, boundary buffering is critical in preventing internal information leaks and external conflict interferences. Boundary buffering enhances information flow management, reduces inadvertent leaks of key information, and mitigates the adverse effects of external environmental changes on idea implementation* (De Vries et al., 2022). *Building on the deductions from Hypothesis* 2b*, informational faultlines through boundary buffering can maximally protect the team from adverse external influences, thus providing a favorable internal environment for idea implementation. Consequently, we propose Hypothesis* 3b*:*

Hypothesis 3b Informational faultlines indirectly positively influence idea implementation through the mediating role of boundary buffering.

Informational faultlines contribute to critical boundary spanning activities, aiding teams in adapting flexibly to external environments and improving work efficiency, which in turn enhances idea implementation (Mell et al., 2022)*. Boundary spanning not only emphasizes mutual assistance and collaboration with teams at similar levels but also focuses on lobbying for support and resource allocation from upper management to achieve the goals of idea implementation* (Marrone et al., 2022). *Additionally, boundary spanning encourages teams to critically assess existing boundaries, foster negotiations to establish mutually recognized boundaries with relevant stakeholders, and actively interact and cooperate with external social network entities to acquire the necessary knowledge, technology, and resources for idea implementation* (De Vries et al., 2022). *Moreover, boundary spanning promotes proactive engagement by teams to bridge and connect different fields, departments, and stakeholders, diminishing the significance of both internal and external boundaries* (Faraj & Yan, 2009). *By accelerating resource and information sharing and circulation, teams can make decisions guided by collective intelligence, thus facilitating rapid conversion and successful idea implementation* (Valtonen et al., 2023). *Building on the discussion in Hypothesis* 2c*, informational faultlines can facilitate boundary spanning through specialized subgroups, thus providing favorable external conditions for idea implementation. Consequently, we propose Hypothesis* 3c*:*

Hypothesis 3c Informational faultlines indirectly positively influence idea implementation through the mediating role of boundary spanning.

2.3.4. The moderating role of paradoxical leadership

Based on the previous discussions, informational faultlines indirectly positively influence idea implementation through the mediation of team boundary work. Furthermore, our study, grounded in the changing others' relationships framework of social network theory, posits that paradoxical leadership moderates the mediating role of team boundary work. Specifically, paradoxical leadership excels at integrating contradictory and interdependent elements, fulfilling members' competitive needs for both unity and diversity, thereby reducing their perception of informational faultlines (Zhang et al., 2022). It also helps members recognize cognitive differences while viewing diverse information as an advantage and establishing connections between differentiated perspectives, thereby strengthening members' sense of identity and cohesion (Batool et al., 2023; Volk et al., 2023). In this context, informational faultlines do not trigger social categorization that could lead to conflicts, competition, or even hostility among subgroups. Instead, they encourage the connection of heterogeneous knowledge subgroups, enhancing members' sense of identity and focusing team resources on promoting team communication, cooperation, and knowledge sharing, thereby actualizing the positive effects of informational faultlines on boundary reinforcement (Feng et al., 2024). Consequently, boundary reinforcement strategies, by fostering team members' awareness of boundaries, enable members to more precisely define their roles and responsibilities and establish closer collaborative relationships with colleagues, jointly dedicating efforts towards idea implementation (Langley et al., 2019). Based on the aforementioned analysis, we propose Hypothesis 4a:

Hypothesis 4a Paradoxical leadership moderates the mediating effect of boundary reinforcement between informational faultlines and idea implementation, such that this mediating effect is stronger under high levels of paradoxical leadership and weaker otherwise.

Our study suggests that paradoxical leadership moderates the impact of informational faultlines on the team's boundary buffering mechanisms (Volk et al., 2023)*. Subsequently, boundary buffering acts as a filtering mechanism for team resources, refining advantageous resources conducive to idea implementation* (Leicht Deobald et al., 2022)*. Specifically, paradoxical leadership maintains both organizational stability and flexibility, effectively recognizing current tasks and environmental conditions, actively filtering out factors that disrupt internal processes to ensure the team rationally focuses on external interactions* (Zhang et al., 2015)*. Moreover, paradoxical leadership shapes the collective strength in the environment, emphasizing not only the acquisition of resources and information from the external environment necessary for completing tasks but also providing guidance and shaping when necessary to enable the team to filter, ignore, and exclude disruptive factors to the task* (Kundi et al., 2023)*. Therefore, under the moderating influence of paradoxical leadership, informational faultlines not only achieve internal boundary reinforcement to ensure team superiority but also external boundary spanning to garner resource support, and establish buffer zones between the internal and external team to ensure rational work* (Valtonen et al., 2023)*. Furthermore, boundary buffering isolates unnecessary external pressures and interruptions, creating a relatively free and relaxed environment for team members, allowing more time and space to think, experiment, and implement new ideas* (Leicht Deobald et al., 2022)*. Based on this analysis, we propose the following hypothesis:*

Hypothesis 4b Paradoxical leadership moderates the mediating effect of boundary buffering between informational faultlines and idea implementation, such that this mediating effect is stronger when paradoxical leadership is high and weaker otherwise.

Similarly, under the moderating influence of paradoxical leadership, informational faultlines can shape the team's social interaction mechanisms with the external environment, namely boundary spanning (Kundi et al., 2023; Mell et al., 2022)*. As a result, boundary spanning as an essential pathway for the team to access vital resources and support, can facilitate idea implementation* (Bertello et al., 2022)*. Specifically, paradoxical leadership shapes the collective strength within the environment, enabling the team to comprehensively consider the complexity of the current tasks and prompting members to perceive that the resources within the informational subgroups are insufficient to resolve current issues* (Zhang et al., 2022)*. This stimulates the team to actively seek necessary resources and support from beyond external boundaries. Moreover, paradoxical leadership can serve as a role model, leading members to actively cross the external boundary networks created by informational faultlines to access resources needed to complete team tasks* (Volk et al., 2023)*. Additionally, during the interaction process with external entities, paradoxical leadership can cultivate paradoxical thinking among employees, enhancing their ability to manage internal team boundaries and external environments* (Zhang et al., 2015)*. At this time, the external boundary social networks constructed by informational faultlines help the team acquire exogenous situational resources necessary for achieving team goals, such as policy support, heterogeneous knowledge resources, or financial support* (Langley et al., 2019; Valtonen et al., 2023)*. Furthermore, social interactions with external stakeholders not only help introduce new ideas, knowledge, and technical support that facilitate idea implementation but also establishing connections with other organizations within the industry can accelerate the spread and adoption of creative ideas* (Leicht Deobald et al., 2022; Perry-Smith & Mannucci, 2017)*. Based on the above analysis, we propose the following Hypothesis* 4c*:*

Hypothesis 4c Paradoxical leadership moderates the mediating effect of boundary spanning between informational faultlines and idea implementation, such that this mediating effect is stronger when paradoxical leadership is high and weaker otherwise.

3. Method

3.1. Participants and procedures

Our study employed a questionnaire survey method based on snowball sampling to collect data samples that met the research criteria. The survey participants were from knowledge-based teams in several high-tech enterprises located in Shaanxi, Shandong, Sichuan, and Hunan, covering team types with innovation capabilities, such as design, finance, and R&D. The geographic distribution of the surveyed enterprises across the eastern, central, and western regions helps mitigate systematic biases stemming from regional economic development levels and cultural differences. The survey process consisted of three phases: a pilot study, the main survey, and questionnaire processing. During the pilot study phase, our research selected R&D teams from three companies to verify the rationality of the questionnaire structure, the clarity of its items, and the accuracy of its content. Taking into account the feedback received from the pilot study, the questionnaire underwent further refinement in order to guarantee the scientific validity and precision of the outcomes derived from the main survey. In the main survey phase, to ensure the standardization of the survey process and the reliability of the data, survey personnel underwent uniform training. Before the start of the main survey, the person in charge provided a detailed explanation of the research background and potential contributions to the management of each enterprise, securing significant attention and support from corporate leaders and HR departments. To ensure data confidentiality, the questionnaires were completed anonymously and solely for academic research purposes. The survey was administered during working hours with the aim of enhancing both the response rate and the quality of the collected data. Furthermore, our study employed a data collection method utilizing two sources—leaders and members—across three waves, with each wave spaced one week apart, to mitigate potential common method bias. At Timepoint 1, team members provided personal information and measures related to paradoxical leadership. At Timepoint 2, team members assessed team boundary reinforcement, buffering, and spanning. At Timepoint 3, team leaders provided basic information about themselves and their teams, such as team size and founding year, and rated the outcome variable of idea implementation. During the questionnaire processing phase, researchers meticulously organized and entered data to ensure its accuracy and completeness. Ultimately, the survey included 97 R&D teams comprising 545 members. After organizing the data and excluding incomplete or incorrectly filled samples, the effective sample consisted of 82 teams with 465 members. The effective response rate for team member questionnaires was 85.49 %, and for leader questionnaires, it was 84.54 %. The final effective sample size of the team was 5.79 members, with 67.28 % being male, an average age of 30.25 years, and 69.52 % holding a bachelor's degree or higher.

3.2. Measures

3.2.1. Informational faultlines

In our paper, the Average Silhouette Width method was employed to calculate the strength of informational faultlines (Meyer & Glenz, 2013). Following the research of Cooper et al. (2014), we selected work tenure, educational specialties, and functional background as individual task-related attributes to construct informational faultlines strength.

3.2.2. Paradoxical leadership

Our research utilized the measurement scale for paradoxical leadership developed by Zhang et al. (2015). The scale comprises 22 items, with a typical item being "The leader demands high performance at work, but is not overly critical.".

3.2.3. Team boundary work

The assessment of team boundary work was conducted using the scale established by Fara and Yan (2009). The boundary reinforcement scale includes four items, such as "We strive to create a clear sense of team identity and purpose." The scale for boundary buffering includes four items, such as "We are able to deflect or absorb external pressures to ensure that team work is not disrupted." The scale for boundary spanning includes four items, such as "Team encourages us to obtain information and resources from outside.".

3.2.4. Idea implementation

Our research applied the measurement scale from Baer (2012) to evaluate the construct of idea implementation. This scale includes three items, with a representative item stating, "Our team's creative ideas have been recognized and are now being implemented.".

3.2.5. Control variables

To examine the distinct effects of informational faultlines on idea implementation, we controlled for informational diversity (Qu & Liu, 2017). Work tenure, as a continuous variable, was measured in terms of diversity using the standard deviation method (Harrison & Klein, 2007). Educational specialization and functional background were evaluated as categorical variables using Blau's index (Collins, 1979). Furthermore, to account for potential confounding factors, the analysis controls for variables such as team tenure, siz, and social faultlines (Yao et al., 2021).

4. Results

4.1. Aggregation

The strength of informational faultlines is determined by calculating the Average Silhouette Width for each individual within the team. Idea implementation is directly assessed by team leaders. Therefore, informational faultlines and idea implementation do not require aggregation tests. Our study measured team boundary work and paradoxical leadership at the individual level, necessitating aggregation tests. The specific results calculated are shown in Table 1, where the values for Rwg, ICC(1), and ICC(2) significantly exceed the thresholds of 0.70, 0.12, and 0.50, respectively (James, 1982). This confirms that the study's sample can be successfully aggregated from the individual level to the team level, thereby providing a solid foundation for subsequent analyses.

Table 1
Results of aggregation analysis.

Variables	R_{wg} mean (>0.7)	ICC(1)> (0.12)	ICC(2)> (0.5)
Boundary reinforcement	0.881	0.356	0.758
Boundary buffering	0.886	0.335	0.741
Boundary spanning	0.901	0.276	0.684
Paradoxical leadership	0.953	0.224	0.621

4.2. Reliability and validity assessment

Table 2 shows that the Cronbach's alpha and composite reliability (CR) values for all measured constructs are above 0.7, indicating that the variables measured by the scales used in this study demonstrate high reliability. Furthermore, the average variance extracted (AVE) for each construct surpasses the 0.5 threshold, and the CR values exceed the recommended 0.7 threshold, verifying their convergent validity. Additionally, as evidenced in Table 4, the square roots of the AVE values are higher than their respective inter-construct correlations, demonstrating robust discriminant validity of the measurements.

In addition, our study employed AMOS 27 to conduct a confirmatory factor analysis (CFA) to assess the discriminant validity among variables and the goodness of fit between the data and the model. Since the informational faultlines strength is calculated from existing data, we were unable to conduct CFA testing on the full model. However, we performed detailed CFA testing on the components of the model that include other core variables involved in our study, with the results presented in Table 3. The baseline model in this study is a five-factor model, comprising paradoxical leadership, boundary reinforcement, boundary buffering, boundary spanning, and idea implementation. Subsequently, we combined variables to form four alternative models. The four-factor model merges paradoxical leadership and boundary reinforcement into one factor based on the baseline model; the three-factor model combines boundary buffering and boundary spanning into one factor, building on the four-factor model; the two-factor model merges all variables except for idea implementation into one factor; and the one-factor model combines all five variables into a single factor. As shown in Table 3, the five-factor baseline model demonstrates a good fit, significantly outperforms the single-factor model. This suggests that these five variables have good discriminant validity, and the data show a high level of fit with the model.

4.3. Descriptive statistics and correlations

Table 4 presents the means, standard deviations, and correlation coefficients of the variables in our sample. The data show that the means and standard deviations are within acceptable ranges, suggesting the absence of significant social desirability bias. Moreover, the correlation coefficients for most key variables align with the hypotheses proposed in our study. For example, informational faultlines are positively correlated with boundary spanning ($r = 0.370$, $p < 0.01$) and idea implementation ($r = 0.223$, $p < 0.05$). In addition, our study further employs the variance inflation factor (VIF) to test for multicollinearity among the variables. The results show that VIF values range from 1.134 to 4.823 across different model tests, indicating that no multicollinearity issue among the variables.

Table 2
Results of validity and reliability testing for variables.

Variables	Cronbach's alpha	CR	AVE
Boundary reinforcement	0.862	0.906	0.708
Boundary buffering	0.826	0.880	0.646
Boundary spanning	0.835	0.886	0.660
Paradoxical leadership	0.982	0.985	0.754
Idea implementation	0.758	0.843	0.644

Table 3
Results of confirmatory factor analysis.

Model	χ2/df	RMSEA	SRMR	TLI	CFI	GFI
Five-Factor Model	2.170	0.076	0.088	0.908	0.930	0.945
Four-Factor Model	3.047	0.109	0.101	0.884	0.835	0.832
Three-Factor Model	4.164	0.128	0.123	0.879	0.816	0.821
Two-Factor Model	5.038	0.133	0.137	0.857	0.801	0.835
Single-Factor Model	5.702	0.144	0.157	0.824	0.778	0.795

4.4. Hypothesis testing

4.4.1. Direct effect test and the impact of the independent variable on mediating variables

Our study proposes that informational faultlines positively influence both idea implementation and boundary work (boundary reinforcement, buffering, and spanning). The results of the regression analysis are detailed in Table 5. After controlling for covariates and including informational faultlines in the model, Model 2 demonstrates a significant positive relationship between informational faultlines and idea implementation (β = 0.223, $p < 0.05$), thus confirming Hypothesis 1. A similar pattern is observed in Models 5, 8, and 11, which show significant positive associations between informational faultlines and boundary reinforcement (β = 0.605, $p < 0.01$), boundary buffering (β = 0.646, $p < 0.01$), and boundary spanning (β = 0.535, $p < 0.05$) respectively. Therefore, Hypotheses 2a, 2b, and 2c are all supported.

4.4.2. Mediation effect test

We employed the PROCESS macro in SPSS 27 to analyze the mediating effects using Model 4, setting the resampling at 10,000 iterations and the confidence interval (CI) at 95 %, incorporating all control variables into the model and the results are presented in Table 6. The indirect effect of informational faultlines on idea implementation through boundary reinforcement was 0.361, 95 % CI [0.161, 0.598], excluding zero, thereby supporting Hypothesis 3a. Similarly, the indirect effect of informational faultlines on idea implementation through boundary buffering was 0.280, 95 % CI [0.079, 0.557], excluding zero, thus supporting Hypothesis 3b. Additionally, the indirect effect of informational faultlines on idea implementation through boundary spanning was 0.160, 95 % CI [0.118, 0.374], excluding zero, thereby validating Hypothesis 3c.

Table 6
Results of bootstrap tests for mediation effects and moderated mediation effects.

Categories	Moderation level	Effect	SE	Boot95%CI
Mediation effects				
Informational faultlines → boundary reinforcement → idea implementation	–	0.361	0.111	[0.161,0.598]
Informational faultlines → boundary buffering → idea implementation	–	0.280	0.122	[0.079,0.557]
Informational faultlines → boundary spanning → idea implementation	–	0.160	0.094	[0.118,0.374]
Moderated mediation effects				
Informational faultlines → boundary reinforcement → idea implementation	−1SD	0.243	0.111	[0.038,0.474]
	+1SD	−0.091	0.036	[-0.180,0.183]
Informational faultlines → boundary buffering → idea implementation	−1SD	−0.002	0.111	[-0.199,0.256]
	+1SD	0.166	0.083	[0.029,0.353]
Informational faultlines → boundary spanning → idea implementation	−1SD	−0.092	0.073	[-0.247,0.040]
	+1SD	0.100	0.062	[0.015,0.229]

Table 4
Descriptive statistics and correlations.

Variable	1	2	3	4	5	6	7	8	9	10	11	12
Team tenure	1											
Team size	0.345**	1										
Tenure diversity	0.111	0.194	1									
Professional diversity	0.132	0.444**	0.147	1								
Functional diversity	0.151	0.488**	0.249*	0.327**	1							
Social faultlines	−0.062	−0.203	0.007	−0.127	−0.074	1						
Informational faultlines	−0.035	−0.141	−0.034	−0.079	−0.070	0.062	1					
Paradoxical leadership	0.111	−0.161	−0.124	−0.119	−0.145	0.096	0.434**	0.803				
Boundary spanning	0.019	−0.152	−0.145	−0.122	−0.183	0.066	0.370**	0.570**	0.814			
Boundary buffering	0.079	−0.154	−0.186	−0.102	−0.168	0.080	0.500**	0.568**	0.524**	0.804		
Boundary reinforcement	−0.065	−0.113	−0.058	−0.016	−0.116	0.038	0.448**	0.561**	0.554**	0.571**	0.812	
Idea implementation	0.014	−0.134	−0.234*	−0.198	−0.180	0.086	0.223*	0.586**	0.384**	0.477**	0.573**	0.868
Mean	7.073	5.793	2.602	0.669	0.680	0.472	0.456	4.951	5.162	5.023	5.071	5.007
SD	1.871	1.783	2.069	0.087	0.087	0.064	0.066	0.611	0.432	0.462	0.513	0.558

Notes: N = 82.* $p < 0.05$ ** $p < 0.01$. The data along the diagonal are highlighted in black to represent the square roots of the AVE values.

Table 5
Direct effects and the impact of the independent variable on mediating variables.

Variables	Idea implementation			Boundary reinforcement			Boundary buffering			Boundary spanning		
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Team tenure	0.072	0.069	0.071	−0.030	−0.037	−0.034	0.154	0.146	0.135	0.081	0.075	0.069
Team size	0.021	0.012	0.014	−0.013	−0.038	−0.023	−0.054	−0.080	−0.073	−0.034	−0.056	−0.051
Tenure diversity	−0.204	−0.193	−0.191	−0.042	−0.014	−0.011	−0.172	−0.142	−0.138	−0.115	−0.090	−0.086
Professional diversity	−0.135	−0.141	−0.137	0.074	0.057	0.049	0.006	−0.012	−0.007	−0.029	−0.044	−0.043
Functional diversity	−0.094	−0.088	−0.085	−0.093	−0.079	−0.068	−0.096	−0.080	−0.074	−0.122	−0.109	−0.093
Social faultlines	0.172	−0.022	−0.019	0.137	−0.192	−0.175	0.174	−0.190	−0.184	0.153	−0.215	−0.207
Informational faultlines		0.223*			0.605**			0.646**			0.535*	
Fau			0.224*			0.448**			0.495**			0.367*
R^2	0.122	0.134	0.131	0.130	0.222	0.207	0.210	0.315	0.287	0.117	0.189	0.164

Notes: N = 82. *$p < 0.05$. **$p < 0.01$.

4.4.3. Moderated mediation test

Our study employed the PROCESS macro to examine the moderated mediation model, specifically utilizing Model 7. The sample size was set to 5,000 with a 95 % CI. As shown in Table 6, under high paradoxical leadership, the indirect effect of the boundary reinforcement mediation path was not significant (β = -0.091, 95 % CI [-0.180, 0.183], including zero). Under low paradoxical leadership, the indirect effect was significantly positive (β = 0.243, 95 % CI [0.038, 0.474], excluding zero), hence Hypothesis 4a was not supported. However, under high paradoxical leadership, the indirect effect of the boundary buffering mediation path was significantly positive (β = 0.166, 95 % CI [0.029, 0.353], excluding zero). Under low paradoxical leadership, the indirect effect was not significant (β = -0.002, 95 % CI [-0.199, 0.256], including zero), thus confirming Hypothesis 4b. Furthermore, under high paradoxical leadership, the indirect effect of the boundary spanning mediation path was significantly positive (β = 0.100, 95 % CI [0.015, 0.229], excluding zero). Under low paradoxical leadership, the indirect effect was not significant (β = -0.092, 95 % CI [-0.247, 0.040], including zero), thereby validating Hypothesis 4c.

4.4.4. Robustness test

To verify the robustness of the regression results, our study adopts the approach of Xi and Qu (2025) by applying the Fau algorithm to reconstruct the informational faultlines strength (denoted as Fau in the model) and re-examine the hypotheses (Xi & Qu, 2025). The results indicate that, compared to the model using the ASW algorithm to calculate the informational faultlines strength, no significant differences were observed, thereby confirming the robustness of the model. The detailed results of the robustness checks are as follows. First, as shown in Models 3, 6, 9, and 12 in Table 5, informational faultlines exhibited significant positive relationships with idea implementation (β = 0.224, $p < 0.05$), boundary reinforcement (β = 0.448, $p < 0.01$), boundary buffering (β = 0.495, $p < 0.01$), and boundary spanning (β = 0.367, $p < 0.05$), thus Hypotheses 1, 2a, 2b, and 2c are once again validated. Second, the robustness test results of the mediating effect in Table 7 indicate that the indirect effect of informational faultlines on idea implementation through boundary reinforcement was 0.265 (95 % CI [0.124, 0.418]); through boundary buffering was 0.240 (95 % CI [0.120, 0.394]); and through boundary spanning was 0.128 (95 % CI [0.026, 0.282]). Therefore, Hypotheses 3a, 3b, and 3c are again validated. Third, the robustness test results of the moderated mediating effect in Table 7 show that paradoxical leadership moderates the indirect effect of informational faultlines on idea implementation through boundary reinforcement, with a moderated mediation index of −0.116 (95 % CI [-0.211, −0.018]). This result did not show significant differences compared to the test using the ASW algorithm to calculate the informational faultlines strength, as previous tests indicated that the indirect effect was significant when paradoxical leadership was low, hence Hypothesis 4a was not supported. Furthermore, the moderated mediation index test also showed that paradoxical leadership significantly moderated the indirect effect of informational faultlines on idea implementation through boundary buffering (moderated mediation index = 0.094, 95 % CI [0.006, 0.189]), and through boundary spanning (moderated mediation index = 0.107, 95 % CI [0.034, 0.190]). Therefore, Hypotheses 4b and 4c are again validated.

Table 7
Robustness tests for mediation and moderated mediation effects.

Mediation effects	Effect	SE	Boot95%CI
Fau → boundary reinforcement → idea implementation	0.265	0.114	[0.124, 0.418]
Fau → boundary buffering → idea implementation	0.240	0.124	[0.120, 0.394]
Fau → boundary spanning → idea implementation	0.128	0.102	[0.026, 0.282]
Moderated mediation effects	Index	SE	Boot95%CI
Fau → boundary reinforcement → idea implementation	−0.116	0.096	[-0.211, −0.018]
Fau → boundary buffering → idea implementation	0.094	0.106	[0.006, 0.189]
Fau → boundary spanning → idea implementation	0.107	0.063	[0.034, 0.190]

5. Discussion and implications

Informational faultlines, by enhancing knowledge diversity and optimizing organizational structures within teams, can facilitate team innovation output (Qu & Liu, 2017). Idea implementation, as a core indicator of team innovation output, directly reflects the team's ability to effectively transform innovative ideas into tangible products or solutions (Kim & Choi, 2022). Therefore, our study, grounded in social network theory, investigates the relationship between informational faultlines and idea implementation for the first time. Our findings indicate that informational faultlines promote idea implementation. This finding aligns with the results of Feng et al. (2024), who demonstrated that informational faultlines enhance team innovation performance, but our study elucidates the specific mechanisms through which informational faultlines facilitate idea implementation. Further research in our study indicates that informational faultlines facilitate idea implementation through the mediating function of team boundary work (boundary spanning, buffering, and reinforcement). This finding corroborates the perspective of social network theory that informational faultlines, as social networks with rich knowledge resources and diversified subgroup structures, can enhance team innovation by effectively managing boundaries to ensure both internal coordination and stability while facilitating the flow of information and sharing of resources (Uzzi, 1997; Zhang & Guler, 2020). This conclusion is consistent with existing research on boundary work, such as Leicht Deobald et al. (2022) and Bertello et al. (2022), who noted that effective boundary work can optimize knowledge flow and resource allocation within teams, reduce conflicts, and enhance team performance. Additionally, our study also demonstrates that paradoxical leadership effectively enhances the mediating impact of boundary buffering and spanning between informational faultlines and idea implementation. This aligns with the majority of existing research on the direct impact of paradoxical leadership on organizational innovation, which suggests that paradoxical leadership enhances innovation outcomes by promoting information and resource integration, thereby improving the team's ability to navigate complex and contradictory situations (Kundi et al., 2023; Wei et al., 2023; Zhang et al., 2022). However, our research also unexpectedly revealed that paradoxical leadership adversely moderates the mediating role of boundary reinforcement in the connection between informational faultlines and idea implementation. This finding is consistent with the latest reviews on paradoxical leadership, which indicate that the continuous balancing of contradictory demands by paradoxical leaders may lead to psychological stress, team conflict, and inconsistent expectations, ultimately having a negative impact on organizational innovation (Batool et al., 2023).

5.1. Theoretical implications

Our study offers the following theoretical contributions. First, it constructs and validates a novel theoretical model grounded in social network theory, exploring the influences of informational faultlines, paradoxical leadership, and team boundary work on idea implementation. This addresses a significant research gap regarding the impact of informational faultlines on idea implementation. Meanwhile, it extends the application of social network theory in the realm of informational faultlines research, offering novel frameworks and insights for deciphering the mechanisms behind the effects of informational faultlines. Innovation defined as the introduction and practical application of novel and potentially valuable ideas, processes, products, or procedures, constitutes a complex array of coherent activities involving not only the stimulation of innovative thinking and generation of new ideas but more crucially, the implementation and execution of these ideas (Perry-Smith & Mannucci, 2017). Existing research on the interplay between informational faultlines and team innovation outputs has primarily focused on the stage of idea generation, largely overlooking the implementation phase. Our study represents an initial exploration of how informational faultlines influence idea implementation. By addressing this overlooked aspect, our research not only contributes to bridging the gap regarding the impact of informational faultlines on idea implementation but also responds the calls from scholars such as Valtonen et al. (2023) and Berg and Yu (2022) for enhanced focus on this area. Furthermore, our study expands the scope of antecedents to idea implementation and the outcomes associated with informational faultlines in this domain.

Second, our study innovatively incorporates team boundary work into the research on the relationship between informational faultlines and idea implementation. It reveals that boundary reinforcement, buffering, and spanning serve as critical mediating mechanisms through which informational faultlines affect idea implementation, thereby expanding the exploration of mediation mechanisms in the path from informational faultlines to team innovative outcomes. Existing research on the mediating mechanisms of informational faultlines typically views teams as closed systems, focusing primarily on internal cognitive, behavioral, or emotional processes (Choi & Sy, 2010; Ma et al., 2022; Qu & Liu, 2017; Zhang & Chen, 2023). This approach largely overlooks the important mediating role of external processes between informational faultlines and team innovation outputs, as well as neglects how team boundary work is influenced by informational faultlines and their impact on idea implementation. Given that idea implementation requires the integration of internal and external resources, team boundary work, which consolidates internal cohesion and acquires external resources, becomes a crucial driving factor for idea implementation. Based on this, our paper explores and tests how team boundary work (boundary spanning, buffering, and reinforcement) mediates the relationship between informational faultlines and idea implementation from an open system perspective. This finding not only reveals the complex path mechanism between informational faultlines and idea implementation but also provides insights and inspiration for exploring the mediating role of boundary work in the study of informational faultlines from a comprehensive, systematic, and in-depth open systems perspective. Moreover, our study extends the research on the consequences of team boundary work and its antecedent factors.

Third, our study reveals the moderating role of paradoxical leadership in the "informational faultlines—team boundary work (boundary spanning, buffering, and reinforcement)—idea implementation" relationship, offering a novel and significant perspective for understanding the boundary conditions under which informational faultlines influence idea implementation. Although existing research has explored the boundary conditions of team faultlines from the perspective of contingent leadership (Kunze & Bruch, 2010; Mitchell & Boyle, 2021; Yao et al., 2021), this perspective employs an "either/or" decision-making logic, emphasizing the match between specific situations and leadership styles, and fails to manage the negative effects of informational faultlines and the seemingly contradictory yet interrelated demands they trigger (Smith & Lewis, 2011). In contrast, paradoxical leadership, which employing a "both/and" thinking mode, underscores the connection and interdependence between contradictions, helping teams to strategically manage informational faultlines and integrate the advantages of both inward and outward boundary work, ultimately fostering idea implementation (Zhang et al., 2015). Therefore, the introduction of paradoxical leadership serves as a critical entry point for addressing the complexities and multiple contradictory demands triggered by informational faultlines in dynamic environments, and offers a comprehensive and precise answer to the inquiries regarding the conditions under which informational faultlines may bolster team boundary work, subsequently affecting the idea implementation. This finding not only furnishes a theoretical foundation for organizations to proactively counteract the adverse consequences of informational faultlines and maximize their beneficial impacts, but it also enhances the research concerning the boundary conditions governing the effects of informational faultlines.

5.2. Practical implications

The primary managerial implications of our study are as follows. First, to effectively manage and leverage the differentiated influences of informational faultlines, organizations should deeply understand and respect the diverse backgrounds of team members while strategically utilizing subgroups that naturally form based on factors such as tenure, functional background, and educational expertise. These subgroups not only facilitate the coordination of team boundary work but also significantly enhance the team' s ability to implement innovative ideas. Therefore, managers should strategically allocate human resources with similar backgrounds to allow the team to maintain a moderate level of diversity while also enabling the formation of subgroups. This configuration strategy increases members' sense of belonging and security, allowing individuals with the same or similar backgrounds to engage in deep exchanges to meet the demands of R&D work for efficient information processing and decision-making. Moreover, when building diverse teams, members should be selected based on characteristics directly relevant to the team tasks to ensure that these traits form multiple complementary subgroups within the team. Simultaneously, HR departments should balance differences among individuals during the recruitment process to avoid introducing isolated individuals who differ significantly in profession or background from existing team members. Such excessive differences may hinder their integration into the team, preventing their unique insights from being understood and supported by other members, and their specialized knowledge from being fully absorbed and utilized by the team.

Second, promoting paradoxical leadership is particularly crucial in teams facing informational faultlines. Paradoxical leadership is especially suitable for managing diverse teams with strong informational faultlines, as it facilitates effective cross-boundary collaboration and boundary buffering strategies, optimizing the team's configuration and utilization of internal and external resources. This not only enhances the team's adaptability to complex situations but also promotes the implementation of creative ideas and the sustainable development of innovation. Therefore, in building and managing teams with high informational faultlines strength, organizations can incorporate the paradoxical characteristics of team leadership behavior into HR management practices such as recruitment and selection, training, performance appraisal, and compensation rewards to enhance the level of paradoxical leadership within team.

5.3. Limitations and further research

This study has some limitations and shortcomings. First, our study employed team leader assessments for evaluating idea implementation. Although the mean and standard deviation of idea implementation in our study were within normal ranges, indicating that social desirability bias was not significant, future research could improve the robustness of findings by combining subjective scales with objective criteria to measure the level of idea implementation. Second, while this study validated its hypotheses using scientific research methods combined with survey data, the applicability of its conclusions to Western contexts or other types of teams remains to be explored. Therefore, future research could expand the survey sample to more multinational corporations, and other national samples to retest the hypotheses of this study and verify whether the findings have broader universality. Third, this study did not focus on social-demographic faultlines but concentrated on informational faultlines. In future research, scholars could consider both social-demographic and informational faultlines in a comprehensive model, using the same variables as contextual factors in the model, to explore whether they have different influences on team boundary work and idea implementation. Fourth, although this study collected data through a multi-source, multi-timepoint questionnaire approach, the time intervals were relatively short. Future research could consider using more timepoints and longer time intervals for data collection.

CRediT authorship contribution statement

Cunhu Xi: Writing – original draft, Data curation, Conceptualization. **Yingqin Zhang:** Supervision, Investigation. **Xiaoqian Qu:** Writing – original draft, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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